

A Counterexample to Conjecture 00000008256: a Maximal Condorcet Domain Above the Stated Bound

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Abstract

The first claim of Conjecture 00000008256 asserts that every maximal Condorcet domain on n alternatives has at most $2^{n-1} n! / 2^{n(n-1)/2}$ linear orders. For $n = 3$ this bound equals 3. The single-peaked domain on three alternatives, consisting of the four orders in which the middle alternative is not ranked last, is a maximal Condorcet domain with 4 orders. The refutation is checked in Lean 4 against Mathlib.

1 The conjecture as stated

The original text of conjecture 00000008256 reads:

Definition: The maximal subdomains of the Condorcet domain for voting paradoxes: the maximal preference domains `cond_max` admitting Condorcet winners. Conjecture: The size of `cond_max` (a maximal Condorcet domain) is at most the Fishburn type explicit bound $2^{n-1} (n!) / 2^{n(n-1)/2}$ (a Fishburn bound formula); the tightness of the bound (domains attaining it) is given by white-list domains (a white-list tight domain law); the count of maximal domains (the number of maximal domains at $n = 4$) is exactly 7 (a four-alternative seven-domain law); and the permutation-symmetric maximization structure of maximal domains is characterized by the lattice of separable preferences (a separable lattice characterization law).

The Chinese version in `conjectures/00000008256.md` writes the bound as $2^{n-1} \cdot (n!) / 2^{n(n-1)/2}$.

2 Reading of the statement

The conjecture is a conjunction of four claims. We refute the first one, so the conjecture is false whatever the other three claims mean.

Fix n alternatives. A *preference* is a strict linear order on the alternatives, and a *domain* is a set D of preferences. A *profile* over D with m voters assigns to each voter $i \in \{1, \dots, m\}$ a preference $\succ_i \in D$; different voters may have the same preference.

Definition 1. Let $(\succ_i)_{i \leq m}$ be a profile.

1. Alternative a *beats* b if more than half of the voters prefer a to b : $\#\{i : a \succ_i b\} > m/2$.
2. a is a *Condorcet winner* if it beats every other alternative.

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3. A domain D admits Condorcet winners if every profile over D with an odd number of voters has a Condorcet winner.
4. D is a maximal Condorcet domain if it admits Condorcet winners and every domain $D' \supseteq D$ that admits Condorcet winners equals D .

Restricting to an odd number of voters is the standard convention: it rules out ties, which otherwise block a strict majority winner even in very small domains (two voters with opposite preferences). The first claim then reads:

$$|D| \leq B(n) := \frac{2^{n-1} n!}{2^{n(n-1)/2}} \quad \text{for every maximal Condorcet domain } D \text{ on } n \text{ alternatives.} \quad (\text{B})$$

If “the size of cond_max” is read instead as the largest size of a Condorcet domain, the claim is refuted by the same example, since that largest size is at least the size of any maximal Condorcet domain.

3 Result

For $n = 3$ we have $B(3) = 2^2 \cdot 6/2^3 = 3$. Write the alternatives as $0, 1, 2$ and a preference as the list of alternatives from best to worst, so that 102 means $1 \succ 0 \succ 2$. Let

$$\mathcal{S} = \{012, 102, 120, 210\}$$

be the orders in which the middle alternative 1 is not ranked last. These are exactly the orders that are single-peaked with respect to the axis $0 < 1 < 2$.

Theorem 2. $\mathcal{S}$ is a maximal Condorcet domain on three alternatives with $|\mathcal{S}| = 4 > 3 = B(3)$. Hence (B) fails, and Conjecture 00000008256 is false.

4 Proof

Lemma 3. $\mathcal{S}$ admits Condorcet winners.

Proof. This is Black’s median voter theorem for three alternatives; we give the direct argument. Take a profile over $\mathcal{S}$ with m voters, m odd. Let N_0 and N_2 be the numbers of voters ranking 0 , respectively 2 , first.

If $N_0 > m/2$, then 0 is a Condorcet winner, since every voter ranking 0 first prefers 0 to each other alternative. Likewise, if $N_2 > m/2$, then 2 is a Condorcet winner.

Otherwise $N_0 \leq m/2$ and $N_2 \leq m/2$, hence $N_0 < m/2$ and $N_2 < m/2$ since m is odd. The only order in $\mathcal{S}$ with 0 on top is 012 , and in the other three orders $102, 120, 210$ the alternative 1 is above 0 . So exactly $m - N_0 > m/2$ voters prefer 1 to 0 . Symmetrically, the only order in $\mathcal{S}$ with 2 on top is 210 , and in $012, 102, 120$ the alternative 1 is above 2 . So $m - N_2 > m/2$ voters prefer 1 to 2 . Hence 1 is a Condorcet winner. $\square$

Lemma 4. $\mathcal{S}$ is a maximal Condorcet domain.

Proof. By Lemma 3, $\mathcal{S}$ admits Condorcet winners. Let $D' \supseteq \mathcal{S}$ admit Condorcet winners, and suppose D' contains an order outside $\mathcal{S}$. The only orders outside $\mathcal{S}$ are 021 and 201 .

- If $021 \in D'$, consider the profile $(021, 210, 102)$ over D' . Here 0 beats 2 (voters 1 and 3), 2 beats 1 (voters 1 and 2), and 1 beats 0 (voters 2 and 3).
- If $201 \in D'$, consider the profile $(201, 012, 120)$ over D' . Here 0 beats 1 (voters 1 and 2), 1 beats 2 (voters 2 and 3), and 2 beats 0 (voters 1 and 3).

In either case the majority relation is a cycle, so every alternative is beaten by some other one and there is no Condorcet winner, a contradiction. Hence $D' = \mathcal{S}$. $\square$

Proof of Theorem 2. By Lemma 4, $\mathcal{S}$ is a maximal Condorcet domain, and $|\mathcal{S}| = 4 > 3 = B(3)$. $\square$

5 Formalization

The Lean 4 project in `lean/` (file `lean/Submission/Basic.lean`) formalizes the definitions above and proves Theorem 2 against Mathlib.

- A preference on `Fin n` is encoded by its ranking `r : Equiv.Perm (Fin n)`, where `r a` is the position of `a` (with 0 the top) and `a` is preferred to `b` iff `r a < r b`. This encodes every strict linear order exactly once.
- `Beats v a b` is `m < 2 * #{i | v i a < v i b}` for a profile `v : Fin m -> Ranking n`; `IsCondorcetWinner`, `AdmitsCondorcetWinners` (all profiles with `Odd m`) and `IsMaximalCondorcetDomain` follow the definition above.
- `SizeBound` is (B) , with the bound computed in $\mathbb{R}$.
- `singlePeaked` is $\mathcal{S}$, defined as the rankings with `r 1  $\neq$  2`; `card_singlePeaked` gives $|\mathcal{S}| = 4$.
- `admitsCondorcetWinners_singlePeaked` is Lemma 3, proved for all odd m by the counting argument above. `cycle_of_not_mem` provides the cyclic three-voter profiles, by `decide`, and `isMaximalCondorcetDomain_singlePeaked` is Lemma 4.
- `conjecture_00000008256_false` is $\neg \text{SizeBound}$, and `conjecture_00000008256_false'` states $\neg(\text{SizeBound} \wedge P \wedge Q \wedge R)$ for arbitrary propositions P, Q, R standing for the other three claims.

The toolchain is `leanprover/lean4:v4.33.1`, Mathlib is pinned to `v4.33.1`, and `lake build` succeeds. The project contains no `sorry`, no `admit` and no `native_decide`, and introduces no axioms: `#print axioms conjecture_00000008256_false` reports exactly `propext`, `Classical.choice` and `Quot.sound`.

6 Remarks

Remark 5. For three alternatives and an odd number of voters, the majority relation is a tournament, and a tournament on three vertices has a Condorcet winner exactly when it is acyclic. So on three alternatives our notion coincides with the other common definition of a Condorcet domain (the majority relation is transitive for every profile with an odd number of voters), and $\mathcal{S}$ is maximal in that sense too.

Remark 6. The bound fails for every $n \geq 3$, although only $n = 3$ is formalized. The single-peaked domain for an axis on n alternatives has 2^{n-1} orders and admits Condorcet winners by Black's theorem [1], and it lies in some maximal Condorcet domain. On the other hand $B(n) < 2^{n-1}$, because $n! < 2^{n(n-1)/2}$ for $n \geq 3$. Moreover $B(n+1)/B(n) = (n+1)/2^{n-1}$, so $B(3) = B(4) = 3$, $B(5) = 15/8$, $B(6) = 45/64$, and $B(n) < 1$ for all $n \geq 6$. Every maximal Condorcet domain is nonempty, because a domain with a single order admits Condorcet winners. So for $n \geq 6$ every maximal Condorcet domain violates (B) .

Remark 7. We do not address the other three claims of the conjecture.

References

- [1] D. Black, *On the rationale of group decision-making*, J. Political Economy 56 (1948), 23–34.